

L11: Robustness of Networks to Random Failures

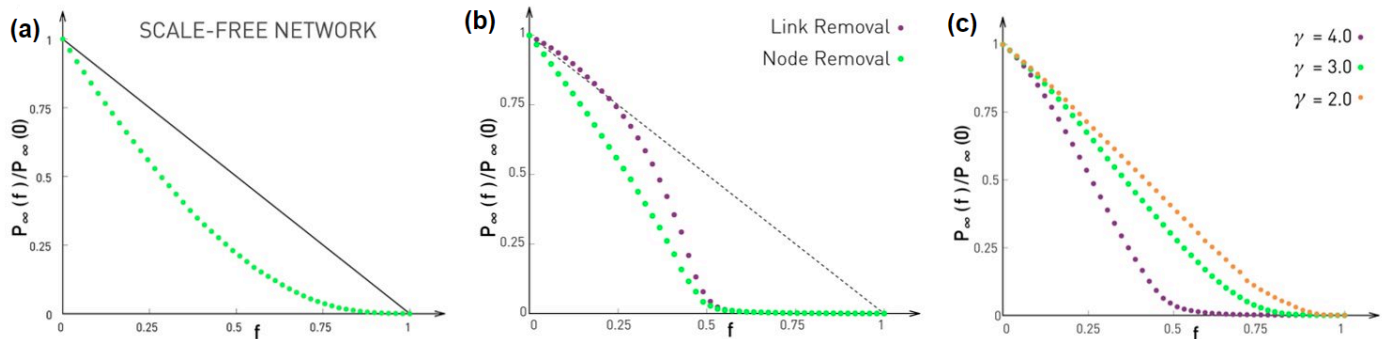


Image Source: **Network Science** by Albert-László Barabási .

We will now present the critical threshold under random node failures. The detailed proof of this result can be found in the textbook (Advanced Topics 8.C). The key points of the proof however are the following:

A. When we remove a fraction f of the nodes, the degree distribution changes. A node that had degree k in the original network, will now have a degree $\kappa \leq k$ with probability

$$\binom{k}{\kappa} f^{k-\kappa} (1-f)^{\kappa}$$

because each of the $k - \kappa$ neighbors of that node is removed with probability f , and there are $\binom{k}{\kappa}$ ways to choose κ from k neighbors.

B. If the probability that a node has degree k at the original network is p_k , the probability that the node has degree κ in the *reduced network* (the network that results after the removal of a fraction f of the nodes) is:

$$p'_{\kappa} = \sum_{k=\kappa}^{\infty} p_k \binom{k}{\kappa} f^{k-\kappa} (1-f)^{\kappa}$$

C. The average degree of the reduced network is a fraction $(1-f)$ of the average degree of the original network:

$$\bar{\kappa} = (1-f) \bar{k}$$

D. Similarly, the second moment of the degree distribution of the reduced network is:

$$\bar{\kappa}^2 = (1-f)^2 \bar{k}^2 + f(1-f) \bar{k}$$

where $\bar{k}^2$ is the second moment of the degree distribution of the original network.

Going back to the Molloy-Reed criterion, the reduced network has a giant connected component if

$$\frac{\bar{\kappa}^2}{\bar{\kappa}} \geq 2$$

Substituting the previous expressions for the first and second moments of the reduced network, we get that the critical threshold for random node failures is

$$f_c = 1 - \frac{1}{\frac{\bar{k}^2}{\bar{k}} - 1}$$

For $G(n,p)$ networks, the degree variance is equal to the average degree, and so $\bar{k}^2 = \bar{k}(\bar{k} + 1)$. This means that the critical threshold for a $G(n,p)$ network is

$$f_{c,G(n,p)} = 1 - \frac{1}{\bar{k}}$$

So, if the average degree of a large $G(n,p)$ network is 2, we expect that the giant connected component of the network will disappear when the fraction of removed nodes exceeds about 50%, as shown in Figure (b).

What happens with power-law networks in which the degree exponent is between 2 and 3? As we saw in Lesson-4, in that case the first moment is finite but the degree variance and the second moment diverge. So, f_c converges to 1, at least asymptotically,

$$f_c, 2 < \gamma \leq 3 \rightarrow 1$$

Figure (a) illustrates this point with simulation results on a power-law network $n=10,000$ nodes, degree exponent $\gamma = 2.5$, and minimum degree $k_{min} = 1$. As n tends to infinity, the green curve will asymptotically converge to zero at $f=1$.

This is a remarkable result that needs further discussion: it means that such networks stay connected in a single component even as we remove almost all their nodes. Intuitively, this happens because networks with diverging degree variance have hubs with very large degree. Even if we remove a large fraction of nodes, it is unlikely that we remove all the hubs from the network, and so the remaining hubs keep the network connected.

The situation is not very different **if we randomly remove links** instead of nodes, as shown in Figure (b). Here, we remove a fraction f of all links in the network. It can be shown that the network's giant connected component disappears at the same critical threshold as in the case of node removals. The visualization in Figure (b) refers to a $G(n,p)$ network in which $n=10,000$ nodes and the average degree is $\bar{k} = 2$. As predicted by the critical threshold equation, the giant component disappears when we remove $1-(1/2)=50\%$ of all edges. For lower values of f , the effect of random node removals is more detrimental than the effect of random link removals (why?).

What happens with power-law networks in which the degree exponent γ is larger than 3? In that case the degree variance is finite, and so we can use the critical threshold equation to calculate the maximum value of f that does not break the network's giant component. Figure (c) shows numerical results of three values of γ – the network size is $n=10,000$ and the minimum node degree $k_{min} = 2$.

Food For Thought

How do you explain that the random removal of a fraction f of nodes causes a larger decrease in the size of the largest component than the random removal of a fraction f of links (when $f < f_c$)?